# Morgan Wang's Portfolio

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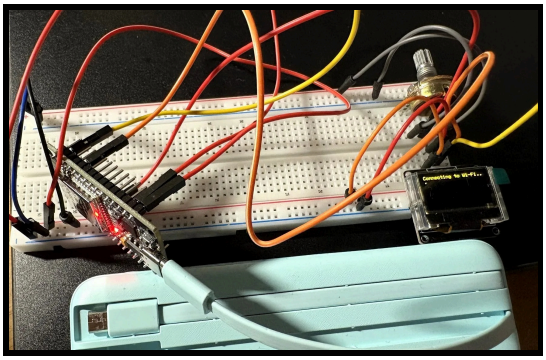
## ESP32 Wireless Telemetry - UW Formula Electric

**What:** The previous UW Formula Electric TCU sent vehicle data over a cellular modem and SIM card, which added a monthly cost and was unreliable at racetracks. Before comparing non-cellular options on paper, I built a small Wi-Fi prototype to get real data on how far a low-cost link can go before data is lost, and whether it fails gradually or in bursts.

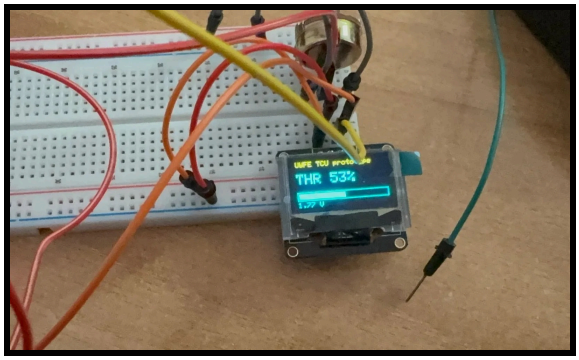
**How:** An ESP32 reads a potentiometer standing in for a throttle sensor, shows live status on an I2C OLED, and sends about 17.5 UDP packets per second. A Python logger on the laptop stores every packet in SQLite. Each packet carries a sequence number, so lost packets show up as gaps. I tested 6 conditions, 3 runs of 30 s each, and used SQL queries and matplotlib to analyze them.

**Result:** Loss stayed at 0.4% or less at most positions up to 15 m, but reached 12% at one 10 m spot and 42.7% behind a closed door. Losses came in bursts of up to 5.8 s, and loss rose sharply once the signal fell below about -80 dBm.

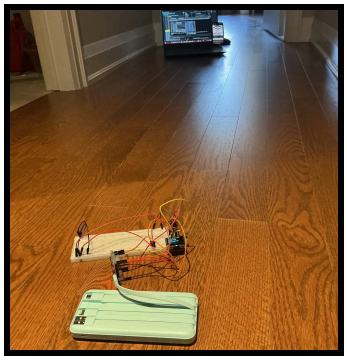
**What it means for the TCU:** Basic Wi-Fi with an on-board antenna is far short of racetrack range. The new TCU needs a radio built for range, such as LoRa or a 900 MHz link, an external antenna mounted high on the car, and a buffer that stores data during dropouts and resends it.



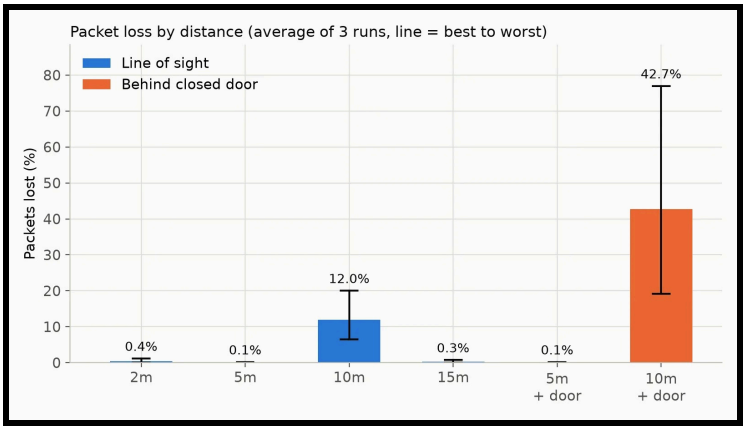
**ESP32 Telemetry Prototype**



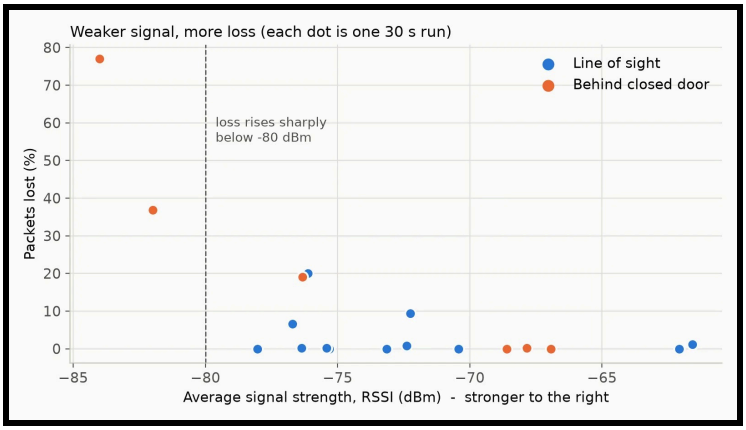

**Throttle on the OLED**



**Range Test at 2 m**




**Packet Loss by Distance**




**Packet Loss vs Signal Strength**

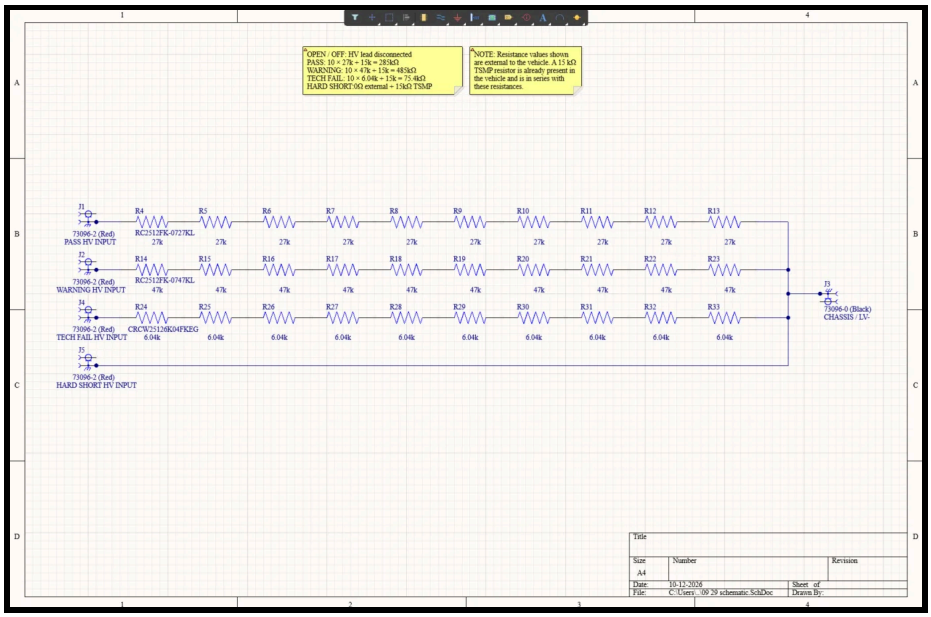
## IMD Test Box - UW Formula Electric

**What:** Developing a passive insulation monitoring fault injection test box for the Formula Electric vehicle, interfacing through the HV+/HV- Tractive System Measuring Points (TSMP) and chassis reference for controlled test conditions.

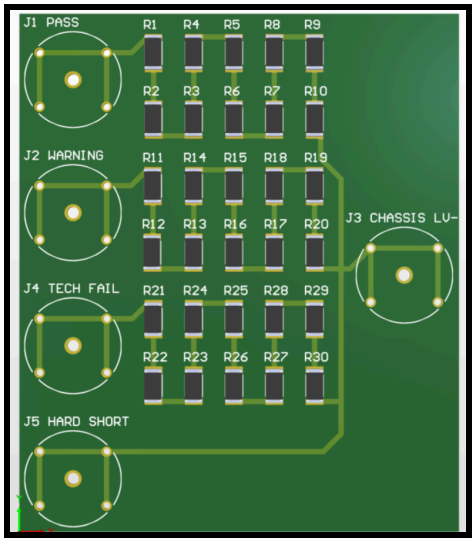
**How:** Created Altium Designer schematics and resistor networks, accounting for the 15 kOhm TSMP protection resistance. No single resistor met both the power and voltage ratings at 588 V, so I split each fault path into a 10-resistor series chain, keeping every 2512 resistor under 0.4 W and 60 V. The test positions, target fault resistances and expected IMD responses come from a chart provided by the electrical team leads, and I designed the box to meet them.

**Result:** Designed resistor networks around the leads' 147 kOhm critical insulation threshold, for target equivalent fault resistances of 285, 485, and 75 kOhm. Planned selection states include PASS, WARNING, TECH FAIL, and a hard short scenario. The design provides a repeatable method to investigate firmware reported warnings and vehicle shutdown responses.

**Next steps:** Currently under review by the electrical team leads. Physical testing on the car will follow.




**Altium Schematic**




**IMD Test Box PCB Render**

| Position | Function | Target total fault (R_fault) | Box resistor (R_box) | Worst case box power (P_box) | Expected IMD response |
|---|---|---|---|---|---|
| 1 | OPEN/OFF | ∞ (disconnected) | Open circuit | 0W | Normal operation (OK_hs high). Safe position for connecting leads |
| 2 | PASS | ≈285kΩ | 270kΩ | 0.85W | Car stays on: Well above 147 kΩ; no false tripping |
| 3 | WARNING | ≈500kΩ | 470kΩ - 485kΩ | 0.48W | CAN Warning: Flags warning status over CAN; car stays on |
| 4 | TECH FAIL | 75kΩ (50% rule margin) | 60kΩ | 3.7W | Mandatory Trip: OK_hs drops to ≤0.2V; SDC must open in ≤30s |
| 5 | HARD SHORT | 15kΩ (sees only R_bpr) | 0Ω (direct wire) | 0W (car BPR takes 23W) | Immediate Trip: Direct HV-to-chassis short simulation |

**Test Conditions and Vehicle Response (chart from the electrical team leads)**

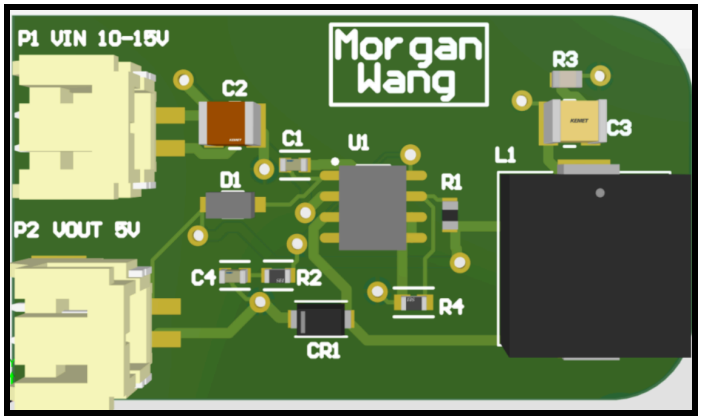
## 5 V DC-DC Buck Converter - Design Task

**What:** Designed a compact 10-15 V input to regulated 5 V, 1 A switching buck converter using the LT1375 for a low voltage power supply design task.

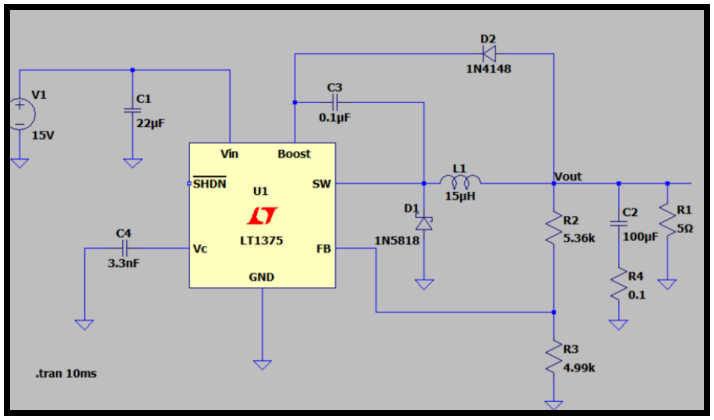
**How:** Selected a 15 uH inductor, Schottky freewheeling diode, input/output capacitors, and resistive feedback components using datasheets and calculations. Simulated switching behavior and component trade offs in LTspice.

**Result:** Reduced simulated output ripple from 45.7 mVpp (v1) to 23.6 mVpp (48.3%, v2) by changing output capacitor ESR from 0.10 to 0.05 Ohm and tuning compensation (1 kOhm / 3.3 nF) at 15 V input and 1 A load.

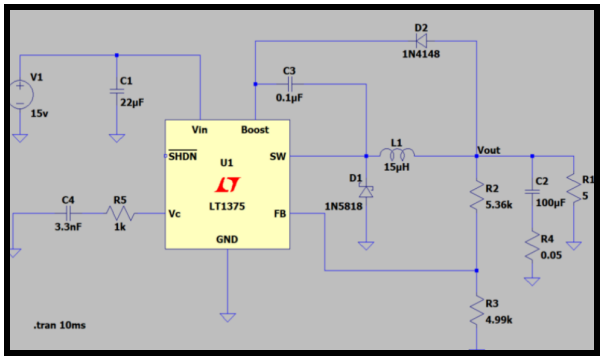
**PCB:** Designed a 41.4 x 23.6 mm two layer SMD layout in Altium Designer with 25 mil power traces, 10 mil signal traces, a bottom layer GND polygon, and via connected returns.



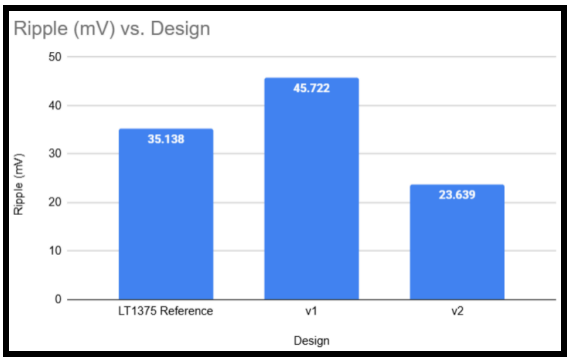

**v2 Buck Converter PCB Render**



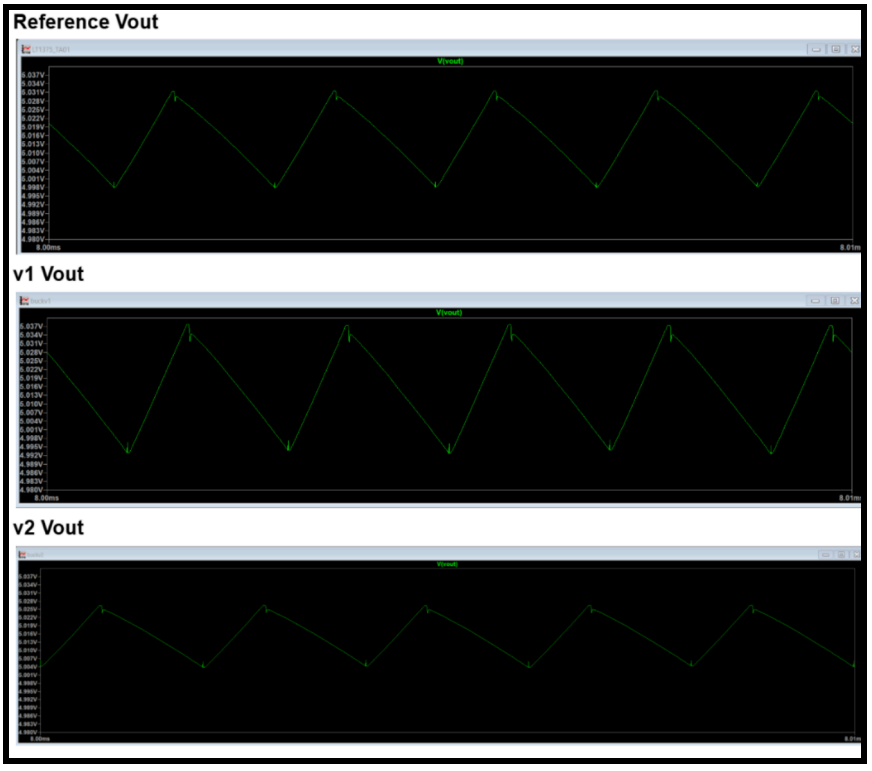

**v1 LTspice Schematic**




**v2 LTspice Schematic**




**Simulated Output Ripple Comparison**



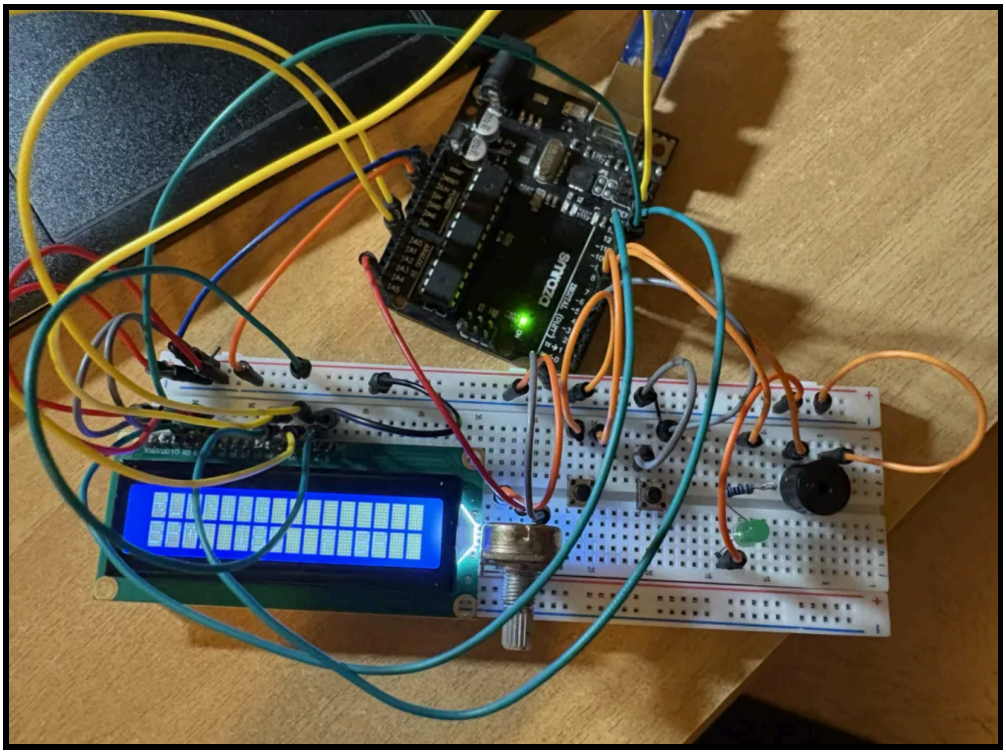

**Output Voltage Comparison**
**(same time interval and y-axis, 4.98–5.04 V)**

## Morgan's Metronome v1.0 - Personal Project

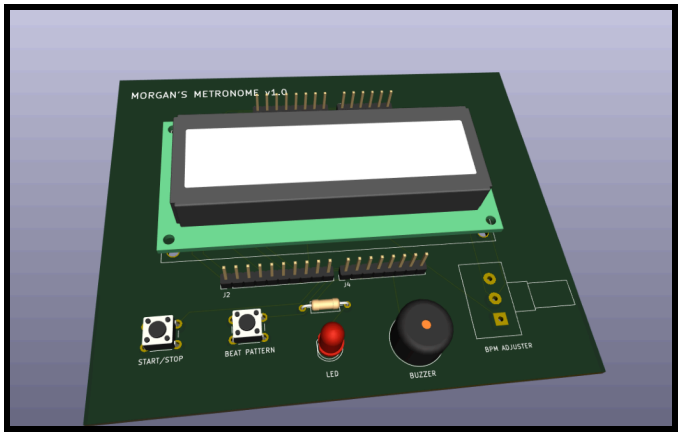
**What:** Built an Arduino UNO based digital metronome after finding myself getting distracted using my phone while practicing saxophone. Designed it to help maintain a consistent rhythm, featuring adjustable 40–180 BPM tempo, selectable beat patterns, pushbutton controls, a potentiometer, and a 16×2 LCD.

**How:** Programmed the timing and user interface in C++. Used millis() based timing to schedule beats so the tempo does not drift; the loop only pauses for the 60 ms beep, so the tempo and button controls stay responsive. Prototyped and debugged the circuit on a breadboard before designing a custom Arduino UNO shield PCB in KiCad to integrate the components.

**Result:** Developed a functional metronome with an audible buzzer and visual LED beat indicators, adjustable tempo, and selectable rhythm patterns. Will order and test the PCB soon.



**Hardware Prototyping on Breadboard**




**KiCad Arduino Shield 3D Render**



**1602A LCD Display, Running at 180 BPM**